| Prompt |
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| # Implementer Prompt - Iteration 01<br><br>Implement only the first backend slice for student management.<br><br>Project context:<br>- Node + TypeScript backend<br>- JSON persistence<br>- current server already boots successfully<br>- keep changes small and localized<br><br>Required scope for this iteration:<br>- create a modular student backend slice<br>- implement GET /api/students<br>- implement POST /api/students<br>- persist students in a JSON file under the data directory |
| # Implementer Prompt - Iteration 02<br><br>Before making any code changes, read and consider these files from the workspace:<br><br>- sistema/.github/copilot-instructions.md<br>- sistema/docs/project-scope.md<br>- sistema/docs/iterations/iteration-02/00-iteration-brief.md<br>- sistema/docs/iterations/iteration-02/01-planner-output.md<br>- sistema/server/src/routes/index.ts<br>- sistema/server/src/modules/students/index.ts<br>- sistema/server/src/modules/students/student.repository.ts<br>- sistema/server/src/modules/students/student.validation.ts<br>- sistema/server/src/shared/types/domain.ts<br><br>Do not edit any file yet. |

# Implementer Prompt - Iteration 03

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-03/00-iteration-brief.md
- sistema/docs/iterations/iteration-03/01-planner-output.md
- sistema/server/src/routes/index.ts
- sistema/server/src/modules/students/index.ts
- sistema/server/src/modules/students/student.repository.ts
- sistema/server/src/modules/students/student.service.ts
- sistema/server/src/modules/students/student.validation.ts
- sistema/server/src/shared/types/domain.ts

# Implementer Prompt - Iteration 04

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-04/00-iteration-brief.md
- sistema/docs/iterations/iteration-04/01-planner-output.md
- sistema/server/package.json
- sistema/server/src/modules/students/index.ts
- sistema/server/src/modules/students/student.service.ts
- sistema/server/src/modules/students/student.repository.ts
- sistema/server/data/students.json

Do not edit any file yet.

# Implementer Prompt - Iteration 05

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-05/00-iteration-brief.md
- sistema/docs/iterations/iteration-05/01-planner-output.md
- sistema/client/src/pages/StudentsPage.tsx
- sistema/client/src/config/api.ts
- sistema/client/src/types/domain.ts
- sistema/client/src/router.tsx
- sistema/client/src/styles.css

Do not edit any file yet.

# Implementer Prompt - Iteration 06

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-06/00-iteration-brief.md
- sistema/docs/iterations/iteration-06/01-planner-output.md
- sistema/client/src/pages/StudentsPage.tsx
- sistema/client/src/config/studentsApi.ts
- sistema/client/src/styles.css

Do not edit any file yet.

First, respond only with:

# Implementer Prompt - Iteration 07

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-07/00-iteration-brief.md
- sistema/docs/iterations/iteration-07/01-planner-output.md
- sistema/server/src/modules/students/student.validation.ts
- sistema/server/src/modules/students/student.service.ts
- sistema/server/src/modules/students/index.ts

Do not edit any file yet.

First, respond only with:

# Implementer Prompt - Iteration 08

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-08/00-iteration-brief.md
- sistema/docs/iterations/iteration-08/01-planner-output.md
- sistema/client/src/pages/StudentsPage.tsx
- sistema/client/src/config/studentsApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 09

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-09/00-iteration-brief.md
- sistema/docs/iterations/iteration-09/01-planner-output.md
- sistema/client/src/pages/StudentsPage.tsx
- sistema/client/src/config/studentsApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 10

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-10/00-iteration-brief.md
- sistema/docs/iterations/iteration-10/01-planner-output.md
- sistema/server/src/modules/classes/index.ts
- sistema/server/src/routes/index.ts
- sistema/server/src/shared/types/domain.ts
- sistema/server/data/README.md

Do not edit any file yet.

# Implementer Prompt - Iteration 11

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-11/00-iteration-brief.md
- sistema/docs/iterations/iteration-11/01-planner-output.md
- sistema/server/src/modules/classes/index.ts
- sistema/server/src/modules/classes/class.validation.ts
- sistema/server/src/modules/classes/class.repository.ts
- sistema/server/src/modules/classes/class.service.ts
- sistema/server/src/shared/types/domain.ts
- sistema/server/data/classes.json

# Implementer Prompt - Iteration 12

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-12/00-iteration-brief.md
- sistema/docs/iterations/iteration-12/01-planner-output.md
- sistema/client/src/pages/ClassesPage.tsx
- sistema/client/src/config/api.ts
- sistema/client/src/router.tsx
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 13

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-13/00-iteration-brief.md
- sistema/docs/iterations/iteration-13/01-planner-output.md
- sistema/client/src/pages/ClassesPage.tsx
- sistema/client/src/config/classesApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 14

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-14/00-iteration-brief.md
- sistema/docs/iterations/iteration-14/01-planner-output.md
- sistema/server/src/modules/assessments/index.ts
- sistema/server/src/routes/index.ts
- sistema/server/src/shared/types/domain.ts
- sistema/server/src/modules/classes/class.repository.ts
- sistema/server/data/classes.json

Do not edit any file yet.

# Implementer Prompt - Iteration 15

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-15/00-iteration-brief.md
- sistema/docs/iterations/iteration-15/01-planner-output.md
- sistema/client/src/pages/ClassesPage.tsx
- sistema/client/src/config/classesApi.ts
- sistema/client/src/config/studentsApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 16

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-16/00-iteration-brief.md
- sistema/docs/iterations/iteration-16/01-planner-output.md
- sistema/docs/iterations/iteration-16/02-implementer-prompt.md
- README.md
- sistema/features
- sistema/server/features/step_definitions
- sistema/server/features/support/hooks.js
- sistema/server/package.json

# Implementer Prompt - Iteration 17

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-17/00-iteration-brief.md
- sistema/docs/iterations/iteration-17/01-planner-output.md
- sistema/client/src/pages/AssessmentsPage.tsx
- sistema/client/src/config/assessmentsApi.ts
- sistema/client/src/config/classesApi.ts
- sistema/client/src/config/studentsApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

# Implementer Prompt - Iteration 18

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-18/00-iteration-brief.md
- sistema/docs/iterations/iteration-18/01-planner-output.md
- sistema/client/src/pages/AssessmentsPage.tsx
- sistema/client/src/config/assessmentsApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 19

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-19/00-iteration-brief.md
- sistema/docs/iterations/iteration-19/01-planner-output.md
- sistema/client/src/pages/AssessmentsPage.tsx
- sistema/client/src/config/assessmentsApi.ts
- sistema/client/src/styles.css
- sistema/client/src/types/domain.ts

Do not edit any file yet.

# Implementer Prompt - Iteration 20

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-20/00-iteration-brief.md
- sistema/docs/iterations/iteration-20/01-planner-output.md
- sistema/server/src/modules/notifications/index.ts
- sistema/server/src/modules/assessments/index.ts
- sistema/server/src/modules/assessments/assessment.service.ts
- sistema/server/src/routes/index.ts
- sistema/server/src/shared/types/domain.ts
- sistema/server/data/README.md

# Implementer Prompt - Iteration 21

Before making any code changes, read and consider these files from the workspace:

- sistema/.github/copilot-instructions.md
- sistema/docs/project-scope.md
- sistema/docs/iterations/iteration-21/00-iteration-brief.md
- sistema/docs/iterations/iteration-21/01-planner-output.md
- README.md
- sistema/server/package.json
- sistema/client/package.json
- sistema/server/features/support/hooks.js
- sistema/server/features/step_definitions/classes-assessments.steps.js
- sistema/server/src/modules/notifications/index.ts
- sistema/client/src/pages/NotificationsPage.tsx

| Model Name and Version - Agent Name - Skills ar | |
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| **Issues** | **Overall result** |
| The implementation worked functionally, but the validation layer initially had semantic issues in its error modeling. A payload-level validation error was modeled as a field-level error, and the validator also included an artificial fallback branch mainly for TypeScript narrowing rather than domain clarity. A small manual adjustment was necessary. | Accepted with manual adjustment |
| The implementation behaved correctly, but the first validation attempt was temporarily confused by pre-existing duplicated records already present in students.json. The duplicate-check logic worked correctly on a clean dataset, but the iteration depended on cleaning legacy duplicated test data before update validation could be interpreted correctly. | Accepted |

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| The implementation behaved correctly and required no meaningful manual code correction after generation. The main remaining concerns are architectural rather than functional: repository path handling still relies on process.cwd(), and JSON-file persistence remains simple and non-transactional. | Accepted |
| The implementation behaved correctly and did not require backend business-logic changes, but the test layer depends on a separately running local API and still inherits the simplicity of JSON-file state management. The suite also emits a Node-version warning in this environment, although execution succeeds. | Accepted |
| The first students frontend slice worked well and looked good, but manual review found two follow-up UX issues: long student values could still stress the list layout, and CPF input was still too permissive in the UI. These issues are better handled in a dedicated corrective frontend-hardening iteration. | Accepted |

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| The implementation behaved correctly and solved the immediate frontend UX issues, but CPF validation is still enforced only in the frontend. This means invalid CPF values could still be sent directly to the backend API outside the UI, so a backend hardening follow-up is still necessary. | Accepted |
| The implementation behaved correctly and kept the rule centralized, but CPF validation is still intentionally basic. It now enforces numeric-only plus exact length 11, but it does not attempt richer CPF business validation beyond that scope. | Accepted |
| The implementation behaved correctly and delivered edit mode cleanly, but the students page is becoming more feature-dense. The current shared form approach still works well, yet future iterations should keep guarding against excessive logic growth inside the page component. | Accepted |

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| The implementation behaved correctly and kept the delete flow well scoped, but the students page now carries create, edit, and delete behavior in one component. The current localized approach still works, yet future iterations should continue watching for excessive UI-state growth. | Accepted |
| The implementation behaved correctly and used the required class shape, but the classes backend is still intentionally basic. It covers only list/create plus payload validation and does not yet include update/delete, enrollment workflows, or assessment logic. | Accepted |
| The implementation behaved correctly and completed the basic classes backend CRUD, but update still uses full replacement semantics and persistence remains simple JSON file I/O. During manual validation, delete required a clean server restart, which appears to have been a local runtime refresh issue rather than a code bug. | Accepted |

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| The implementation behaved correctly and aligned the classes page with the backend contract, but the classes frontend is still intentionally basic. It currently covers only list/create and does not yet include edit/delete or deeper class flows. | Accepted |
| The implementation behaved correctly and completed the basic classes frontend flow, but the classes page now concentrates create, edit, and delete behavior in one component. The current localized approach still works, yet future iterations should keep guarding against excessive UI-state and stylesheet growth. | Accepted |
| The implementation behaved correctly and stayed within scope, but successful manual validation required using a class with actual enrolled student ids. The persistence model is intentionally simple and the classes repository now holds a small amount of assessment-specific support logic. | Accepted |

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| The implementation behaved correctly and completed the missing student enrollment flow, but the classes page now holds more local UI state and the shared stylesheet continues to grow. The current solution is still coherent and localized, but future iterations should keep avoiding excessive page-level complexity. | Accepted |
| A implementação ficou correta e útil, mas a suíte ainda dependia de a API estar rodando separadamente e a iteração ainda não cobria notifications. Também permaneceu a simplicidade de persistência em JSON, exigindo hooks de reset bem controlados para evitar vazamento de estado entre cenários. | Accepted |
| A implementação funcionou para leitura e visualização da matriz, mas ainda deixou um gap real no fluxo de primeiro lançamento: turmas com alunos matriculados e sem goals salvos caíam em um estado vazio pouco útil. Isso não quebrou a iteração, mas exigiu uma correção posterior. | Accepted with follow-up |

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| A edição célula-a-célula foi implementada corretamente e com bom feedback visual, mas o fluxo ainda dependia de goals já presentes na matriz. Como a tela continuava fraca para o primeiro lançamento em turmas sem avaliações anteriores, a iteração precisou de follow-up corretivo. | Accepted with follow-up |
| The implementation behaved correctly and stayed within scope. The main tradeoffs are the additional state branching in the assessments page and the decision to hardcode the first-use goals (`Requirements`, `Tests`) in the frontend. This is acceptable for the experiment now, but could become a limitation only if later requirements demanded configurable goals. | Accepted |
| The implementation behaved correctly and stayed within scope. The main tradeoffs are that dispatch remains manually triggered and email delivery is still development-oriented for local/demo safety. The Notifications page also depends on combining notifications data with student/class data to display friendly labels. | Accepted |

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| A iteração ficou bem focada em entrega final e não desestabilizou o sistema. Os principais caveats remanescentes são esperados: dispatch manual por design, email development-safe/log-based e dependência de filesystem persistente em hospedagens se quiser retenção de JSON. A revisão do colega também ficou pendente por decisão explícita. | Accepted |

**nd Agents**

| Overall sentiment |
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| Useful for a small backend slice with clear scope, but still required close review for semantic quality and clean validation design. |
| Good result for a medium-sized backend slice. The agent respected scope, improved modularity with a service layer, and implemented the required HTTP behaviors with less manual adjustment than in Iteration 01. |

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| Good result for a focused backend slice. The agent respected the scope, preserved modularity, and extended the students backend cleanly without introducing unnecessary changes. |
| Good result for a test-infrastructure iteration. The agent respected scope, kept the setup simple, and created readable acceptance scenarios with a practical reset strategy. |
| Good result for a first frontend slice. The agent delivered a clearly improved and functional students page with clean visual organization, but a small corrective pass is still needed for stronger UX resilience. |

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| Good corrective iteration. The agent stayed scoped, preserved the visual quality of the page, and resolved the manual-review UX issues without overengineering the solution. |
| Good backend hardening iteration. The agent stayed narrowly scoped, changed only the right layer, and aligned backend validation with the frontend behavior without unnecessary refactoring. |
| Good frontend iteration. The agent stayed scoped, preserved the current visual quality, and added edit mode without breaking the existing create flow. |

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| Good frontend iteration. The agent preserved the current visual style, added delete cleanly, and kept the full students flow coherent without unnecessary complexity. |
| Good backend iteration. The agent stayed scoped, followed the modular pattern already established in students, and introduced the classes slice cleanly without drifting into broader functionality. |
| Good backend iteration. The agent stayed scoped, completed the missing CRUD operations cleanly, and preserved the modular structure of the classes module. |

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| Good frontend iteration. The agent stayed scoped, preserved visual consistency with the students page, and added the first functional classes UI cleanly. |
| Good frontend iteration. The agent preserved the current visual style, added edit and delete cleanly, and kept the classes flow coherent without unnecessary complexity. |
| Good backend iteration. The agent stayed disciplined, implemented the first useful assessments slice, and preserved a simple persistence model without overengineering. |

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| Good frontend iteration. The agent reused the existing students API cleanly, added enrollment without overengineering, and kept the classes page behavior consistent. |
| Boa iteração de testes de aceitação backend. O agente expandiu a infraestrutura existente sem alterar comportamento de produto e manteve os cenários legíveis e determinísticos. |
| Boa primeira fatia do frontend de assessments. A tela passou a carregar turmas, alunos e a matriz, mas ainda não representava de forma completa o caso de uso inicial sem avaliações prévias. |

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| Boa iteração de frontend. O agente adicionou persistência célula-a-célula de forma disciplinada, mas isso ainda não bastava para fechar completamente o gerenciamento de assessments. |
| Good corrective iteration. The agent fixed a real functional gap in the assessments workflow without over-expanding scope, and the page now behaves much more like a usable management screen. |
| Very good iteration. The backend notifications requirement was implemented correctly, and the added visibility layer solved the demonstration gap without requiring a separate iteration. |

Forte iteração final de entrega. Ela melhorou muito a prontidão para avaliação ao fechar a cobertura backend de notifications, fortalecer o README e formalizar o checklist de entrega.

| Observations |
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| The agent respected the requested scope and produced a reasonably modular result with routing, validation, repository logic, and JSON persistence separated. Manual validation confirmed that GET and POST worked correctly and that invalid payloads were rejected. The main quality issue was not functionality, but the design of the validation response shape. This should be constrained more explicitly in future prompts. |
| The agent handled this iteration better than the previous one because the prompt required planning first and explicitly encouraged a service layer when the module grew. Duplicate checks and update rules were centralized more cleanly. The main caveat was not code failure, but confusion caused by previously duplicated test records in the JSON file. Future prompts should continue to enforce thin routes, shared validation, and centralized conflict logic. |

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| This iteration went more smoothly than the earlier ones because the prompt constrained the change set tightly and the students module already had a service/repository structure to extend. The delete behavior was implemented cleanly with correct status codes and correct persistence updates. The next prompts should continue using the same discipline of thin routes and localized changes. |
| This iteration was successful because the prompts constrained the solution to minimal Cucumber infrastructure and discouraged unnecessary complexity. The use of JavaScript step definitions and deterministic reset hooks kept the setup understandable. The main lesson is that stable acceptance tests for JSON-backed systems require explicit state control. |
| The agent handled the visual requirement reasonably well once the prompt explicitly required a polished result. The best outcome came from combining functional scope with aesthetic constraints in the prompt. Manual review remains essential because overflow and input-hardening issues were only visible after real interaction with the page. |

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| This iteration worked well because the prompt was tightly constrained around two concrete review findings: input hardening and layout resilience. The combination of simple input sanitization and targeted CSS was enough to fix the visible problems quickly. The next important step is to align backend validation with the improved frontend behavior. |
| This iteration worked well because the requirement was simple, explicit, and strongly constrained toward one shared validation point. The result improved system integrity by ensuring invalid CPF values can no longer bypass the frontend and enter through direct API calls. |
| This iteration worked well because the prompt explicitly required reuse of the existing form and strong visual edit-mode cues. That prevented unnecessary duplication and kept the page coherent. The next step should complete the students frontend flow with delete functionality. |

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| This iteration worked well because the prompt explicitly separated delete from confirmation-modal complexity and required correct interaction with edit mode. The result completed the core students frontend workflow while keeping the page visually consistent. |
| This iteration worked well because the required class shape and scope boundaries were stated very explicitly. That prevented field-name drift and helped keep the implementation localized. The next step should complete the classes backend with update/delete and then move toward class-related frontend flows. |
| This iteration worked well because the scope was tightly limited to update/delete and required consistent 404 handling. Reusing the existing validator helped keep the implementation simple. Manual testing also reinforced that local dev-server restarts can matter when validating fresh backend changes. |

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| This iteration worked well because the payload requirements were made explicit and the visual direction was tied to the already-working students page. That kept the slice small, coherent, and easy to validate. The next step should complete the classes frontend flow with edit/delete actions. |
| This iteration worked well because the payload requirements were made explicit and the visual direction was tied to the already-working students page. That kept the slice small, coherent, and easy to validate. The next step should complete the classes frontend flow with edit/delete actions. |
| This iteration worked well because the scope was tightly limited to one read flow and one single-cell update flow. Requiring strict validation for class, student, goal, and concept kept the behavior predictable. The resulting structure is simple enough to support the future assessments UI without prematurely expanding into notifications or analytics. |

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| This iteration worked well because the prompt explicitly forced reuse of the existing students API and asked for a simple checkbox-based enrollment flow. That kept the implementation small, predictable, and easy to validate. With this slice complete, the classes frontend now supports the enrolled-student flow needed before building the assessments UI. |
| A iteração 16 fechou a cobertura backend de classes e assessments com Cucumber/Gherkin, adicionando cenários de CRUD de turmas e dos fluxos principais de assessments. O principal aprendizado foi que, num sistema baseado em JSON, a confiabilidade dos testes depende diretamente do reset explícito de dados por cenário. |
| A iteração 17 entregou a base correta do frontend de assessments em modo leitura: seleção de turma, resolução de studentId para nome e montagem da matriz. O problema descoberto depois foi de completude do fluxo, não de quebra técnica da implementação. |

A iteração 18 introduziu o PUT de assessments no frontend, com seletor restrito a MANA/MPA/MA, bloqueio durante save e feedback de sucesso/erro. Mais tarde ficou claro que o maior gap remanescente não era o save em si, e sim a usabilidade do primeiro preenchimento, resolvida depois na iteração 19.

This iteration was important because it removed the dependence on preexisting stored goals in the backend just to make the assessments page usable. That makes the assessments workflow practically complete enough to support the next required feature: daily consolidated notifications by email.

This iteration became stronger after the follow-up UI visibility layer was added. Without that page, notifications would only be visible through logs and JSON files. With the Notifications page in place, the feature is now both implemented and demonstrable in the running application.

A iteração 21 adicionou cobertura de aceitação backend para notifications, endureceu o reset determinístico incluindo notifications.json, reescreveu o README como guia de avaliação e registrou o checklist final de entrega. O sistema terminou tecnicamente pronto para commit, deploy e demonstração.